

```
In [1]: import pandas as pd
```

```
In [2]: books = pd.read_csv("allbacks.csv")
        books.head()
```

```
Out[2]: <style scoped> .dataframe tbody tr th:only-of-type {
        vertical-align: middle; } .dataframe tbody tr th {
        vertical-align: top; } .dataframe thead th { text-align:
        right; } </style> <thead> <th></th>
        <th>rownames</th> <th>volume</th>
        <th>area</th> <th>weight</th> <th>cover</th>
        </thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
        <th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

1	885	382	800	hb
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2	1016	468	950	hb
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3	1125	387	1050	hb
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4	239	371	350	hb
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5	701	371	750	hb
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```
In [3]: books["cover"].sort_values().reset_index(d
```

```
Out[3]: 0      hb
        1      hb
        2      hb
        3      hb
        4      hb
        5      hb
        6      hb
        7      pb
        8      pb
        9      pb
       10      pb
       11      pb
       12      pb
       13      pb
       14      pb
Name: cover, dtype: str
```

<p>the Median in (pb) which is the value of number 8th</p>

```
In [4]: sorted_covers = books["cover"].sort_values
        median_cover = sorted_covers.iloc[len(sort
        print("Median cover type:", median_cover)
```

Median cover type: pb

```
In [5]: mean_weight = books["weight"].mean()
        print("Mean weight of all books:", mean_we
```

Mean weight of all books: 706.6666666666666

<p>The mean weight of all books is 706.67 grams</p>

```
In [6]: mean_by_cover = books.groupby("cover")["we
        print(mean_by_cover)
```

```
cover
hb      796.428571
pb      628.125000
Name: weight, dtype: float64
```

The mean of hb weighs about 796.43 grams, while the mean of pb weighs about 628.13 grams.

The hardcover books are about 168.30 grams heavier than the softcover, which suggests that the cover type is associated with a difference in book weight in this inventory.

```
In [7]: print("Median cover type:", median_cover)
        print("Mean weight of all books:", mean_weight)
        print("Mean hardcover weight:", mean_by_cover_type["hardcover"])
        print("Mean paperback weight:", mean_by_cover_type["paperback"])
```

```
Median cover type: pb
Mean weight of all books: 706.6666666666666
Mean hardcover weight: 796.4285714285714
Mean paperback weight: 628.125
```

```
In [ ]:
```